Pavan Kalyan Venigalla

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SUMMARY

Data Analyst with over **3 years of experience** delivering data-driven insights, automation, and real-time analytics solutions across **banking, investment, healthcare, and retail domains.** Accomplished in designing **end-to-end data pipelines**, building **predictive models**, and developing **interactive dashboards** that improve decision-making and business performance. Proficient in **Python, SQL, Power BI, Tableau, Spark, and AWS**, with hands-on experience in **streaming analytics, data governance, and regulatory reporting (Basel III, OCC, HIPAA).** Demonstrated expertise at collaborating with cross-functional stakeholders to translate business goals into actionable analytics strategies, optimize data accuracy, and enhance operational efficiency in **global enterprise environments**.

SKILLS

Data Analytics & Modeling: Python (Pandas, NumPy, Scikit-learn, PySpark MLlib), SQL, Statistical Modeling, Predictive Analytics, Time Series Forecasting, Data Wrangling, Feature Engineering
Big Data & Streaming: Apache Spark, Kafka, Spark Streaming, AWS Redshift, Snowflake, AWS S3, Airflow, Data Lake Architecture
Business Intelligence & Visualization: Power BI, Tableau, Excel (Pivot, Power Query), KPI Dashboards, Data Storytelling, Performance Benchmarking
Cloud & Automation: AWS Lambda, AWS Glue, Azure Data Factory, ETL/ELT Automation, Data Pipeline Orchestration
Databases & Storage: SQL Server, Redshift, Snowflake, MySQL, PostgreSQL, SAP S/4HANA (Reporting Data), FHIR-compliant Data Structures
Regulatory & Domain Expertise: Financial Risk Analytics (Basel III, CCAR), Fraud Detection, Investment Portfolio Analytics, Claims Analytics, Supply Chain Forecasting, HIPAA Compliance, Data Governance
Collaboration & Agile Tools: JIRA, Confluence, Git, Agile/Scrum, SDLC Documentation, Cross-functional Stakeholder Collaboration

EXPERIENCE

Northern Trust — **Feb 2025 – Present**
Data Analyst — *USA*

- Designed and deployed a real-time fraud detection pipeline leveraging Kafka, Spark Streaming, and AWS Redshift, enabling continuous monitoring of high-volume transactions and improving fraud detection latency by 40%.
- Developed anomaly detection and behavioral analytics models using Python (Scikit-learn, PySpark MLlib) to identify suspicious transaction patterns, reducing fraudulent losses by 22% across credit and fund transfer operations.
- Implemented automated alerting mechanisms integrated with Power BI and AWS SNS, enhancing incident response efficiency and reducing average fraud investigation time by 35%.
- Engineered a scalable data ingestion and transformation framework to process market and transactional data from Snowflake, SQL Server, and AWS S3, ensuring reliable, audit-ready datasets for downstream analytics.
- Automated portfolio performance tracking and benchmarking workflows using Python and AWS Lambda, enabling daily computation of KPIs like Sharpe Ratio, Alpha, and tracking error for institutional portfolios worth $15B+.
- Developed Tableau dashboards visualizing asset allocation, benchmark deviation, and sector exposure, improving data transparency and reducing manual reconciliation workloads by 50%.
- Partnered with risk, compliance, and investment strategy teams to align analytics outputs with Basel III and OCC regulatory standards, driving data accuracy and governance improvements of 25% across reporting systems.

Hexaware Technologies — **May 2020 - Nov 2022**
Data Analyst — *India*

- Optimized credit risk forecasting for a global banking client by designing predictive models in Python (Scikit-learn) and integrating real-time data from Oracle Financials and Bloomberg APIs, improving loan default prediction accuracy by 38% and reducing manual review time by 40%.
- Engineered an automated ETL pipeline using SQL and Airflow to consolidate multi-source finance data, enhancing reporting speed and ensuring 99% data accuracy for monthly regulatory submissions.
- Built interactive Power BI dashboards tracking liquidity, capital adequacy, and NPA ratios, empowering senior executives with self-service analytics and cutting report generation time from 2 days to 3 hours.
- Led claims data analysis initiative for a healthcare insurer, developing fraud detection and readmission prediction models using Python and Tableau, which reduced false claims by 22% and improved patient outcome tracking across 50K+ records.
- Applied FHIR and HIPAA-compliant data modeling to unify structured and unstructured claims data, increasing system interoperability and ensuring 100% compliance with regulatory standards.
- Implemented AI-driven demand forecasting models (ARIMA, Prophet) across retail product categories using Python, boosting forecast accuracy by 35% and enabling optimized procurement cycles.
- Delivered real-time supply chain visibility dashboards integrating SAP S/4HANA and POS data, allowing stakeholders to monitor inventory turnover and vendor performance, reducing overstock by 18% and lead times by 12%.

EDUCATION

University of North Texas — **Dec 2024**
Masters in Computer Science — *USA*

NRI Institute of Technology — **Apr 2022**
B Tech in Computer Science — *India*

CERTIFICATION

- Career Essentials in Data Analysis – Microsoft
- Structured Query Language (SQL) – Coursera
- Data Visualization – LinkedIn Learning